



PTT PHENOL COMPANY
LIMITED

PROCESS SPECIFICATION

ACID AROMATICS SUMP PIT PUMP (P-2320)

Docu. No. : 14780-8120-PS-P2320
Project No. : 120117
Rev. No. : Z1
Date : MAY 10, 2016
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TITLE :

ACID AROMATICS SUMP PIT PUMP

(P-2320)

DISCIPLINE : **PROCESS**
DOCUMENT NO. : **14780-8120-PS-P2320**
PROJECT NAME : **PTT Phenol Train II Project**
PROJECT NO. : **120117**
OWNER : **PTT Phenol Company Limited (PPCL)**

Z1	MAY 10, 2016	AS BUILT	S.W.Park	J.S.Chung	-	S.W.Lee
F1	MAY 28, 2014	FINAL ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
A2	FEB 25, 2014	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
A1	OCT 11, 2013	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
O1	JUL 30, 2013	COMMENT ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
REV. NO.	DATE	DESCRIPTION	PREP'N	REVIEW	QA REVIEW	APPROVAL
REV. NO.	PREPARATION DEPT.	PREPARATION	REVIEW	QA REVIEW	APPROVAL	
Z1	PROCESS	S.W.Park	J.S.Chung	-	S.W.Lee	
		MAY 10, 2016	MAY 10, 2016	-	MAY 10, 2016	

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Job Number 120117 Item Number P-2320 Purchase Order Number _____ Date _____
Req./Spec. Number _____ Inquiry Number _____ By _____

Note : ☐ Indicates Information Completed by Purchaser ☐ by Manufacturer ☒ by Purchaser or Manufacturer

☐ For PTT PHENOL COMPANY LIMITED ☐ Unit CDN

☐ Site MAP TA PHUT, THAILAND ☐ Service ACID AROMATICS SUMP PIT PUMP

☐ No. Req. ONE (1) ☐ Pump Size _____ ☒ Type VERTICAL(MANUFACTURER'S STANDARD)

☒ Manufacturer ☐ Model _____ ☒ Serial Number _____

☐ GENERAL

☐ Number Motor Driven _____ ☐ Number Turbine Driven _____ ☐ Gearbox Item Number _____

☐ Motor Item Number _____ ☐ Turbine Item Number _____ ☐ Gearbox Provided By _____

☐ Motor Provided By _____ ☐ Turbine Provided By _____ ☐ Gearbox Mounted By _____

☐ Motor Mounted By _____ ☐ Turbine Mounted By _____

☐ OPERATING CONDITIONS

	Rated	Max.	Normal	Min.
Capacity (m ³ /h)	5		5	
Suction Pressure (kg/cm ² g)	0			
Discharge Pressure (kg/cm ² g)	1.0 (Note8)			
Differential Pressure (kg/cm ² g)	0.99			
Differential Head (m)	10.0			
Hydraulic Power (kW)				

At Designated Capacity

	Rated	Max.	Normal	Min.
Operating Time (hr./yr.)				
NPSH Available (m) (Note12)			7.91(Note7)	

System Design

☒ Stand Alone Operation ☐ Parallel Operation

☐ Series Operation with Item Number _____ (1 operation)

Suction Pressure Min./Max. _____ / 0.400 (kg/cm² g)

Service

☐ Continuous ☒ Intermittent (Starts/Day) _____

System Control Method

☐ Speed ☐ Flow ☐ Level ☐ Temperature

☐ Pressure ☐ Pipe Friction Resistance Only

☐ PUMPED FLUID

Pumped Fluid WATER

	Rated	Max.	Normal	Min.
Pumping Temperature (°C)			38	

At Designated Temperature

	Rated	Max.	Normal	Min.
Specific Gravity		0.99		
Vapor Pressure (kg/cm ² a)		0.07		
Viscosity (cP)		0.68		
Specific Heat (kJ/kg°C)				

Initial Boiling Point _____ (°C) @ _____ (kg/cm² a)

Liquid ☐ Hazardous ☒ Flammable

☐ Other _____

Corrosion/Erosion Caused By _____

% Solid _____ Max. Particle Size _____ (mm)

Notes :

1. Centrifugal pump shut-off head shall not exceed 125% of specified differential head inclusive of all tolerances.

2. Flowrate excludes any design allowance. No allowance for spillback is included or required.

3. Possibility of auto refrigeration : No

4. Liquid pumped : Flammable(Note 11)

5. Pump suction strainer shall be provided by vendor.

6. Process design pressure is 3.5 kg/cm² g and design temperature is 120 °C.

7. NPSH(a) is based on pump center line to pit BTL.

8. Discharge pressure is considered at pump discharge flange.

9. P-9108S shall be warehouse spare for P-2320.

10. Service for X-2321 Acid aromatics sump

11. Pump liquid is considered as flammable due to potential of oil spillage area of service.

12. 1 m margin for NPSH(a) to be considered.

13. Pump sealing requirement shall be single mechanical Seal.

14. Available voltage for induction motors :

a) Non-process motors up to 0.35 kW: AC 220 V ± 5, 1 Phase, 50 Hz,

b) Motors 0.35 kW up to 159 kW : AC 380 V, 3 Phase, 50 Hz

c) Motors 160 kW and over : AC 6.6 kV, 3 Phase, 50 Hz



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Job Number 120117	Item Number _____	Purchase Order Number _____	Date _____																								
Req./Spec. Number _____	/	Inquiry Number _____	By _____																								
☐ MECHANICAL DATA		☐ COUPLING BETWEEN PUMP AND DRIVER																									
Impellar Type ☐ Closed ☐ Open ☐ Semi-Open ☐ Other _____		Manufacturer _____ Type _____ Size _____ Model _____ Spacer Length (mm) _____																									
Casing Mounting ☐ Foot ☐ Centerline ☐ Vertical Inline ☐ Close-Coupled		Coupling Guard Type ☐ Manufacturer's Standard ☐ Baseplate Mounted ☐ Non-spark Coupling Guard Required																									
Bearings ☐ Bearing Manufacturer _____ ☐ Radial Bearing Type _____ No. _____ ☐ Thrust Bearing Type _____ No. _____ ☐ Bearing Isolators _____ ☐ Shaft Stiffness Ratio (L^3/D^4) _____		Remarks _____ _____ _____																									
Lubrication ☐ Flood ☐ Pure Mist ☐ Shielded (Grease) ☐ Grease ☐ Pure Mist ☐ Sealed (Grease) ☐ Constant Level Oiler Required ☐ Housing Vent Required ☐ Magnetic Drain Plug in Housing Required ☒ Oil Cooler Required ☐ Seal Spray Guard Required ☒ Oil Viscosity ISO Grade _____ ☐ Other _____		BASEPLATE Remarks ☐ Grounded ☐ Free Standing ☒ Centerline of Pump to Still Bottom _____ (mm) ☐ Vertical Pump Case Support Bracket Design ☐ PIP Standard RESP002 (Data Sheet Attached) ☐ Manufacturer's Standard Remarks _____																									
Nozzle Connections ☐ Size ☐ Rating ☐ Facing		PAINT AND SHIPMENT PREPARATION																									
<table border="1" style="width:100%; border-collapse: collapse;"> <tr> <td style="width:30%;">Suction</td> <td style="width:30%; text-align: center;">#150</td> <td style="width:40%; text-align: center;">RF</td> </tr> <tr> <td>Discharge</td> <td style="text-align: center;">#150</td> <td style="text-align: center;">RF</td> </tr> </table>		Suction	#150	RF	Discharge	#150	RF	<table border="1" style="width:100%; border-collapse: collapse;"> <tr> <td style="width:50%;">Pump</td> <td style="width:50%;">Baseplate</td> </tr> <tr> <td>☐ Manufacturer's Standard</td> <td>☐ Manufacturer's Standard</td> </tr> <tr> <td>☐ Other _____</td> <td>☐ Other _____</td> </tr> </table>		Pump	Baseplate	☐ Manufacturer's Standard	☐ Manufacturer's Standard	☐ Other _____	☐ Other _____												
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☐ Other _____	☐ Other _____																										
Aux. Case Connection ☐ Drain Required ☒ Size _____ (mm) ☐ Threaded ☐ Welded and Flanged		Shipment ☐ Domestic ☐ Export ☐ Export Boxing Required Number of Months of Storage _____ Total Weight _____ (kg)																									
☐ MATERIALS		INSPECTION AND TESTING																									
Material Class Code: _____		☐ Final Inspection Required ☐ Days Notification Req'd _____																									
Casing Cast Iron		<table border="1" style="width:100%; border-collapse: collapse;"> <tr> <th style="width:30%;">Test</th> <th style="width:20%;">Non-Witnessed</th> <th style="width:20%;">Witnessed</th> <th style="width:30%;">Certificate</th> </tr> <tr> <td>Hydrostatic</td> <td style="text-align: center;">☐</td> <td style="text-align: center;">☐</td> <td style="text-align: center;">☐</td> </tr> <tr> <td>Performance</td> <td style="text-align: center;">☐</td> <td style="text-align: center;">☐</td> <td style="text-align: center;">☐</td> </tr> <tr> <td>NPSHR</td> <td style="text-align: center;">☐</td> <td style="text-align: center;">☐</td> <td style="text-align: center;">☐</td> </tr> <tr> <td>Vibration</td> <td style="text-align: center;">☐</td> <td style="text-align: center;">☐</td> <td style="text-align: center;">☐</td> </tr> <tr> <td>Other</td> <td style="text-align: center;">☐</td> <td style="text-align: center;">☐</td> <td style="text-align: center;">☐</td> </tr> </table>		Test	Non-Witnessed	Witnessed	Certificate	Hydrostatic	☐	☐	☐	Performance	☐	☐	☐	NPSHR	☐	☐	☐	Vibration	☐	☐	☐	Other	☐	☐	☐
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Other	☐	☐	☐																								
Impellar _____ Cover _____ Shaft _____ Shaft Sleeve _____ Baseplate _____ Casing Gasket _____ Impellar Gasket _____ Casing Fasteners _____ Bearing Housing _____ Bearing Housing Adapter _____ Bearing Housing End Seals _____ Coupling Guard _____ Mechanical Seal Gland _____ Mechanical Seal Fasteners _____		☐ Dismantle and Inspect After Test ☐ Casting Repair Procedure Approval Required Material Certification Required ☐ Casing ☐ Cover ☐ Impeller ☐ Shaft ☐ Other _____																									
		Inspection Required for Connection Welds ☐ Manufacturer's Standard ☐ Visual Inspection																									
		Inspection Required for Castings ☐ Manufacturer's Standard ☐ Visual Inspection ☐ Other _____																									
☐ DRIVER		MANUFACTURER DOCUMENTATION REQUIREMENTS																									
Horsepower Rating _____ (kW) Speed _____ (rpm) Driver HP Selected or Max S.G. _____ & Max. Visc. _____ (cP) Remarks _____		☐ For Vendor Data Requirements Refer to: _____ Remarks _____																									
Notes (Continued) : 15. Available voltage for motor space heater is a) Heater load < 3 kW : 220 V, 1 Phase AC b) Heater load ≥ 3 kW: 380 V, 3- Phase AC 16. Noise level shall not exceed 85 dB(A). 17. Deleted. 18. Lubrication system is applied to high voltage motors' and all pumps' bearings except for vertical and diaphragm pumps.																											

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